Effects of Central Bank Intervention on the Interbank Market During the Subprime Crisis

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We explore whether central bank intervention improves liquidity in the interbank market during the current subprime crisis with unique trade and quote data from the e-MID, the only regulated electronic interbank market in the world. Central bank intervention consistently creates greater uncertainty in the interbank market. Prior to the crisis, the cover-to-bid ratio effectively conveys good and bad news from the central bank, but this link is broken during the crisis, suggesting that standard (and special) interventions that do not specifically target interbank asymmetric information fail to improve market liquidity. Our results suggest that the central bank should release stress tests for individual banks, provide interbank loan guarantees, or engage in direct asset purchases rather than simply providing more capital when counterparty risk poses systemic risk to the interbank market. (*JEL* G01, G21, G28)

Beginning in August 2007, the subprime crisis created serious liquidity problems for the interbank market. Banks, unsure about the depth of the problems on other banks' balance sheets, were simply unwilling to lend to each other without substantial accommodations for counterparty risks. To counteract the resulting lack of liquidity in the interbank market, central banks began to proactively provide liquidity to the system through various means and methods. In fact, the European Central Bank (ECB) made numerous adjustments to and increased the frequency of traditional open market operations while adding

For helpful comments and discussions we thank Matthew Spiegel, the editor, an anonymous referee, Kirsten Anderson, Paolo Angelini, Bahattin Buyuksahin, Alain Chaboud, Michael Fleming, Craig Furfine, Andrei Kirilenko, Fabiola Ravazzolo, Michel Robe, Tano Santos, Chiara Scotti, Clara Vega, and participants at the Yale School of Management-RFS Financial Crisis Conference, the LACEA-LAMES 2009 meeting in Buenos Aires, and the IMF Monetary and Capital Markets Seminar series. We would like to thank e-MID for providing the data. Send correspondence to Celso Brunetti, Carey Business School, Johns Hopkins University, 100 International Drive, Baltimore, MD 21202; telephone (410) 516-8656. E-mail: celsob@jhu.edu.

doi:10.1093/rfs/hhq123 Advance Access publication November 8, 2010

special operations in the attempt to boost market liquidity, reduce liquidity risk, and/or improve funding liquidity.[1] In this article, we specifically study the effects of central bank intervention on the interbank market, examining liquidity, spreads, and volume both prior to and during the ongoing subprime crisis.[2]

We employ a unique dataset that reports trades and quotes of the only electronic, regulated interbank market in the world: the e-MID S.p.A. (or Mercato Interbancario dei Depositi). The e-MID is the reference marketplace to trade liquidity in Europe, and as the first and only electronic market for interbank deposits and overnight indexed swaps in the euro area, e-MID has offered multilateral trading since 1990. The euro money market is the second largest in the world, and the active e-MID euro overnight market offers a comprehensive view of interbank lending both before and during the recent crisis.

We collect comprehensive e-MID data for a pre-crisis period from January 2, 2006, through August 8, 2007, and a crisis period from August 9, 2007, through April 1, 2008. Acharya and Merrouche (2009) document August 9 as a structural break in the interbank market, when suspended redemptions from three money market funds by BNP Paribas generated a cascade of negative announcements from other banks. We examine both ordinary and special interventions by the ECB during these periods to assess how central bank interventions affect interbank market liquidity.

We find that ECB interventions are quickly incorporated into both the conditional mean and the conditional standard deviation of prices, spreads, and volume during the pre-crisis and crisis periods. To further investigate these results, we compute impact responses. The various types of central bank interventions affect market prices unevenly in both pre-crisis and crisis periods. During the pre-crisis period, central bank interventions commonly reduce interbank spreads and increase trading volume as intended. During the crisis, however, neither Main Refinancing Operations (MRO) nor Long Term Refinancing Operations (LTRO) are consistently effective in reducing prices or in reducing interbank market uncertainty. During the crisis, in fact, interventions are commonly accompanied by increased spreads and lower trading volume. Furthermore, liquidity suppliers initiate fewer trades on the ask side during the crisis—evidence that liquidity suppliers become more passive.

More consistently, ECB interventions generally increase the conditional volatility of price changes, spreads, and volume, suggesting that interventions may convey important information to the market. The increase in conditional

[1] Other adjustments include variously narrowing/widening the facilities corridor, extending the list of eligible collateral, offering a fixed-rate tender offer with full allotments, adjusting deposit and lending rates, and conducting one-year LTROs as fixed-rate tenders. Brunnermeier and Pedersen (2009) explicitly model market and funding liquidity.

[2] Hayek (1945) and Rochet and Tirole (1996) demonstrate that interbank markets provide important information aggregation, price discovery, and peer-monitoring roles.

volatility is more prevalent during the crisis, suggesting that this information may be consistently viewed as "bad news" that increases volatility in the interbank market.[3] To explore this possibility, we compare ECB interventions with other significant news events in both pre-crisis and crisis periods. During the crisis, other news events significantly affect price changes, spreads, and volatility, but these effects commonly revert to prior levels within 15 minutes. Conversely, price changes, spreads, volume, and volatility immediately react to central bank interventions but largely do not revert, suggesting that interventions do not simply convey information to the interbank market—some other effect likely intercedes.

We explore whether this other effect is consistently bad news, utilizing the fact that the ECB chooses the level of intervention necessary after bids are received. This cover-to-bid ratio is, in fact, similar across pre-crisis and crisis periods, so we utilize the ratio as a proxy for good and bad news for each ECB intervention. We demonstrate that good and bad intervention news elicits differential responses in price changes, spreads, and volume prior to the crisis (as expected), but no such differences exist during the crisis. We find that even relatively good news increases prices, price change volatility, and spread volatility and fails to improve spreads or volume during the crisis period. These results suggest that crowding-out effects (Heider, Hoerova, and Holthausen 2009) dominate the intervention news effects during the crisis, indicating that the intervention process, whether standard or special, is not effective when liquidity problems stem from asymmetric information between banks.

Our empirical design mitigates the possibility that interventions do not normatively improve liquidity but might, in fact, result in greater liquidity than otherwise might occur in the absence of intervention. In our tests, interbank reactions are isolated to very short-term horizons, within minutes of the intervention. This design also creates a caveat for our results. Although we effectively capture the crowding-out effects in the short run, longer-term effects can be only inferred from our tests. To the extent that adverse changes to spreads and volume represent an impediment to the efficient conduit of information into prices, however, the crowding-out effects we document are likely robust.

While the debate on the effectiveness of central bank measures to counter the recent crisis remains unresolved, our results generally concur with those of Taylor and Williams (2009) and Stroebel and Taylor (2009), who find that U.S. Federal Reserve interventions were not effective. Conversely, McAndrews, Sarkar, and Wang (2008) and Fleming, Hrung, and Keane (2010) show that the Fed's Term Auction Facility and Term Securities Lending Facility reduced LIBOR-OIS (London Interbank Offer Rate-Overnight Indexed Swap) spreads and repo spreads between Treasury and less liquid collateral, respectively.

[3] We thank the editor for prompting us to address this important issue.

Our findings have important policy implications about how central banks address liquidity crises. While Brunnermeier and Pedersen (2009) model the link between market and funding liquidity, our results speak largely to the goal of market liquidity. Consistent with Holmstrom and Tirole (1996), Sundaresan and Wang (2009) show that central bank actions can improve market liquidity when anticipating known liquidity events exogenous to the interbank market. Our findings that direct liquidity provision has limited effects during the current crisis suggest that more nuanced central bank intervention is merited when counterparty risk creates liquidity problems. To this point, the fact that the European Union began to release stress test results for individual banks (directly addressing the asymmetric information problem) on October 2, 2009, is a step in the right direction.[4] Given the prospect that the central bank potentially crowds out private liquidity, we recommend that central banks also pursue alternatives like interbank loan guarantees or direct asset purchases (Blanchard 2009).

Our findings make several contributions to the existing literature and are consistent with Acharya and Merrouche's (2009) evidence of precautionary liquidity hoarding in the UK market. Hamilton (1996), building on Poole (1968), develops an econometric model to mimic the forecasting exercise of the Fed in providing commercial bank liquidity with shocks representing the forecasting error of the central bank. We represent these shocks with the cover-to-bid ratio and supply imbalances. The breakdown between the interbank market and the cover-to-bid ratio during the crisis suggests that the ECB either did not fully grasp the extent of liquidity hoarding or consistently underfunded aggregate bids during the crisis.[5]

Our evidence that central bank intervention increases volatility in the interbank market supports the studies by Gorton (2009) and Heider, Hoerova, and Holthausen (2009), who find that asymmetric information plays an important role in the current credit crunch. As bank accounts deteriorate, increased counterparty risk leads to adverse selection problems that may provoke interbank market failure. Schwarz (2009), in fact, shows that a third of the e-MID spread increase during the crisis represents counterparty risk.

Finally, our article relates to the literature on how central bank intervention affects interbank markets. Allen, Carletti, and Gale (2008) and Acharya, Gromb, and Yorulmazer (2008) model central bank liquidity provision with incomplete markets and with market power, respectively.[6] With incomplete

[4] The U.S. Fed first took the step to release stress test results for individual U.S. banks in the spring of 2009 in a clear nod to address information asymmetries among banks.

[5] In the vast literature on the impact of news on financial markets (e.g., Andersen, Bollerslev, Diebold, and Vega 2002, 2007; Ehrmann and Fratzscher 2005a, 2005b; and Fleming and Remolona 1999, among others), we are the first to address the effects of ECB news announcements on high-frequency interbank data. Consistent with prior work, we find that news is quickly incorporated into prices and volatilities.

[6] See also Freixas and Jorge (2008).

markets, the central bank restores liquidity when the interbank market reaches a suboptimal equilibrium (with higher price volatility). With market power, central bank intervention should reduce the cost of borrowing, the volatility of borrowing costs, transaction costs (bid-ask spreads), and spread volatility while bolstering trading volume. Our article empirically tests these hypotheses.

The rest of the article is organized as follows. Section 1 describes the e-MID market and the ECB interventions. Section 2 contains details on the high-frequency transactions and quote data. Section 3 describes the econometric approach, and Section 4 contains results. In Section 5, we analyze impact responses, and we conclude in Section 6.

1. The e-MID Market and European Central Bank Interventions

The e-MID company manages the only interbank unsecured deposit market on an electronic platform for the euro system. Established in 1990 as an initiative of the Bank of Italy, the e-MID initially served the euro-interbank market but rapidly expanded to the U.S. dollar, the British pound, and the Polish zloty. During the pre-crisis period, the e-MID held an estimated 17–22% market share in all euro interbank transactions, with plausibly diminished market share during the crisis. The e-MID is open to all the banks admitted to operate in the European interbank market, while non-European banks may access the market via their European branches.

About 200 commercial banks based in Europe participate in the euro interbank market, the most developed e-MID segment. Treasury traders may access the market, post quotes (bid, ask, or both), and execute trades in full transparency. Interbank deposits range from overnight (1 day) to 2 years in duration, with overnight contracts representing about 90% of the total volume during our sample period.

The main function of the euro interbank market is to allow participant banks to effectively reallocate deposit imbalances among themselves. Most institutions recur to various ECB facilities to manage gross liquidity needs and access the interbank market to cope with daily liquidity shocks. Interbank transactions may occur via brokers, phone, or the e-MID platform.

The market operates simply, with each trader observing the real-time book of quotes, which displays the amount and interest rate for participant bids and offers. “Best Quotes” (highest bid and lowest ask) are displayed at the top of the book, with others displayed by descending terms. Trading begins daily at 8:00 a.m. European Central Time (ECT) and ends at 6:00 p.m. ECT. Each trader may decide to initiate a transaction with any counterparty present in the book, depending traditionally on posted terms, credit line availability, and mutual trading history. When a trader *hits* the quote displayed on the book, the system allows both traders to negotiate terms (both the amount and the interest rate). Once the transaction is executed, it is automatically processed into the payment system, and the real-time book automatically updates.

All banks operating a reserve account with the ECB are subject to an average minimum reserve requirement computed according to the quantity and quality of the bank's assets. A Main Refinancing Period (MRP), typically 4 or 5 weeks, determines the period over which this average is calculated. At the end of every MRP, a new minimum average amount is determined by the ECB.

During each MRP, the ECB employs a wide set of tools to fulfill its monetary policy objectives and maintain the target rate at the determined level. In order to do so, the ECB periodically injects (or absorbs) reserves into (or out of) the European banking system. These operations take the form of repurchase agreement (repo) contracts between the ECB and commercial banks. For example, in a monetary injection, the central bank receives titles from commercial banks and in turn credits the relative amount on their deposits with the ECB. At the expiration of the repo, the operation is then reverted from commercial banks to the ECB. For the majority of the operations, the end of a repurchase agreement coincides with the beginning of a new agreement, so that the net liquidity change is the difference between two consecutive repos submitted to the system.

Of the three regular ECB monetary policy operations, MROs are the most important in that they are used to signal the stance of monetary policy each week, as described in Figure 1. The ECB announces its benchmark amount every Friday, sometimes revising it the following Monday. Each bank bids for funds with an offered rate between Monday and Tuesday morning. MROs are usually held every Tuesday, when liquidity is allotted through a variable rate auction mechanism.[7] The central bank assigns reserves starting with those banks that pledged a higher rate before satisfying those with a lower bid rate.

LTROs are characterized by a maturity of 3 months and are offered infrequently, usually at the end of each calendar month. Although LTROs are not considered primarily important for monetary policy, the ECB recurred to

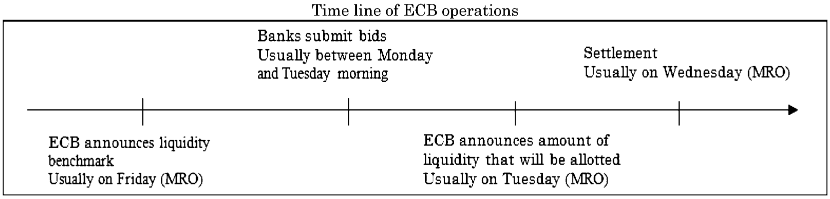

Figure 1
Typical time line of ECB operations
MROs are typically conducted weekly. LTROs and FT operations are also conducted with the same sequence but at different times and days.

7 Effective October 15, 2008, weekly MROs are executed at fixed rather than variable rates.

LTROs more frequently during the crisis, adding *special* operations not conducted at the traditional dates. We demonstrate below that these largely unexpected special operations convey a greater element of surprise to the market.

Fine-tuning (FT) operations are usually held on the last day of a reserve maintenance period. The central bank fine-tunes the market to keep interbank market prices close to the policy rate in the event of unforeseen changes in the ECB's estimate of autonomous liquidity factors.[8] These operations usually last one day and may either inject or absorb liquidity. We are able to identify many *special* FTs during the crisis as well as according to whether they (i) last longer than one day; (ii) have been executed on a day other than the final day of an MRP; (iii) tender an amount that is considerably larger than the standard amount of the FT operations.

The ECB also executes monetary policy by guiding market expectations about the reference rate, the overnight rate set as a target for interbank transactions. The ECB monetary policy committee usually meets on the first Thursday of every month to decide whether to change the reference rate. The decision is subsequently announced publicly to guide expectations for smoothing the size of injections and absorptions of liquidity. As Demiralp and Jorda (2002) show, setting correct expectations (announcement effects) reduces the need for and size of central bank interventions.

2. High-frequency Data: e-MID

For our empirical analysis, we first collect high-frequency quote and trade data from e-MID. The high-frequency e-MID data are proprietary and contain detailed information on transactions (day, time stamp, price, and volume) and the full order book of quotes spanning 573 trading days from January 2, 2006, through April 1, 2008. We analyze quotes and transactions when the market is particularly active: from 8:30 a.m. ECT until 5:00 p.m. ECT.[9] We construct all ECB interventions during this sample period from data on the official ECB website and then classify each according to the type of intervention based on direct conversations with ECB officials. To benchmark e-MID market reactions to other newsworthy events, we also collect news items from Factiva, Lexis, and Westlaw.

We take care with our high-frequency data to avoid the contamination of microstructure noise in our measurements. For optimal sampling frequency, we use both volatility signature plots (Andersen, Bollerslev, Diebold, and Labys 1999) and the procedure in Bandi and Russell (2006) to minimize the effect

8 Autonomous liquidity factors represent those factors affecting liquidity in the euro system that are not directly under control of the central bank.

9 Angelini, Nobili, and Picillo (2009) also use e-MID data during the crisis and find that higher spreads between secured and unsecured interbank rates are explained by higher risk aversion, borrowers' creditworthiness, and seasonal factors. Gabrieli (2009) shows that counterparty and default risk play important roles in explaining e-MID rate volatility during the crisis.

of microstructure noise. Since both procedures select a sampling frequency just above 4 minutes, we adopt a conservative approach and analyze 5-minute intervals.

From the full book of quotes, we compute representative spreads for each 5-minute interval across the trading day. Each quote may be valid during one or more intervals. We compute the spread at a given time *t* as the difference between the bid and ask prices, weighted by their posted size and by time so that we capture the time-weighted and size-weighted average spread over each 5-minute interval.[10] This procedure allows us to take into account the depth of the full order book and its time-varying structure, since the best bid-ask spreads might not fully represent market conditions.[11]

Figure 2 shows daily median bid-ask spreads. During the pre-crisis period, seasonality in spreads relates to the weekly MROs and monthly FTs. As shown, bid-ask spreads increase markedly after August 9, 2007, as does the volatility of the spread.

Figure 3 shows the intraday pattern of the median bid-ask spread for the pre-crisis and crisis periods separately. It is evident from Figure 3 that spreads have increased dramatically since August 9, 2007. In fact, the average post-crisis spread is approximately four times the pre-crisis spread. In addition, spreads during the crisis are widely variable during the trading day, contrasting with the relatively stable intraday spreads found pre-crisis. During the crisis, spreads

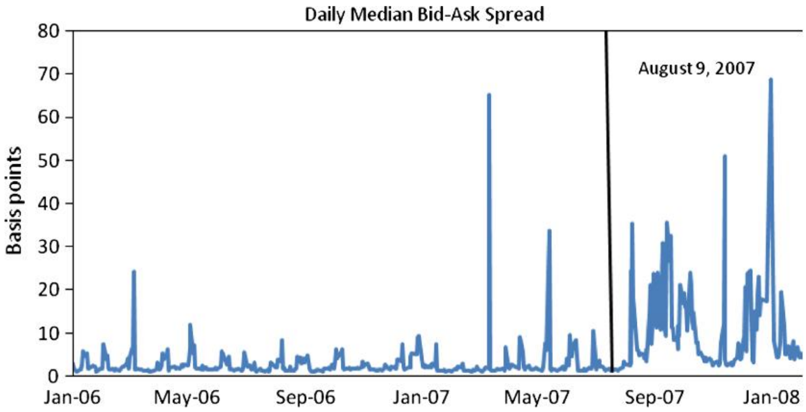

Figure 2
Daily median bid-ask spread
This figure plots the daily median bid-ask spread on e-MID. For each 5-minute interval, the spread is a time-volume weighted average of the order book. The vertical line indicates the beginning of the crisis period.

10 For details, see the Appendix.

11 We also consider the bid-ask spread based on the best quotes and obtain qualitatively similar results to those reported below.

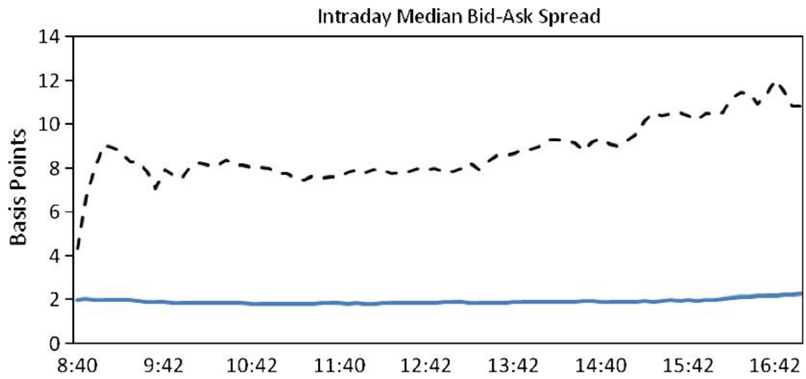

Figure 3
Intraday median bid-ask spread
This figure plots the intraday median bid-ask spread on e-MID. The solid line refers to the pre-crisis period, while the broken line refers to the crisis period.

are relatively low at the opening of the market, increase sharply during the first few minutes of trading, and generally continue to rise for the remainder of the trading day.

Figure 4 presents the median daily depth for bids and offers in e-MID. As shown, the size of the bid generally exceeds the size of the offer throughout the sample period, suggesting more willingness to buy than to sell in this market. During the crisis, both bid and ask depth appear to shrink, a clear indication of increased reluctance to expose orders on e-MID. As Figure 5 shows, the median intraday depth drops by approximately one-third during the crisis.

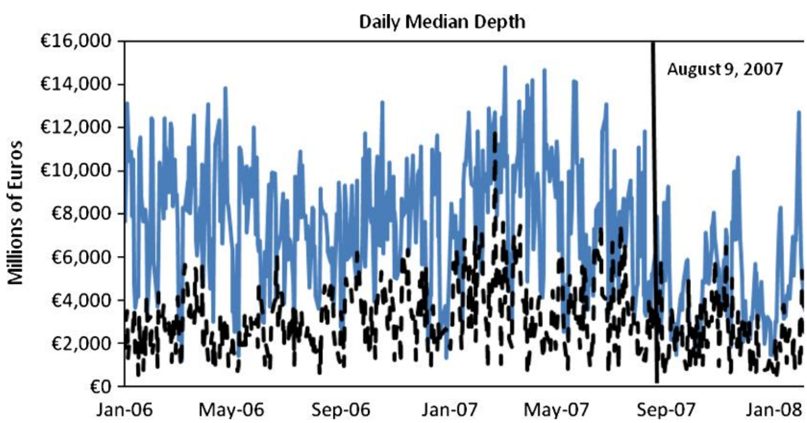

Figure 4
Daily median depth
This figure plots the daily median depth on e-MID. The solid line refers to the daily median bid, while the broken line refers to the daily median ask. The vertical line indicates the beginning of the crisis period.

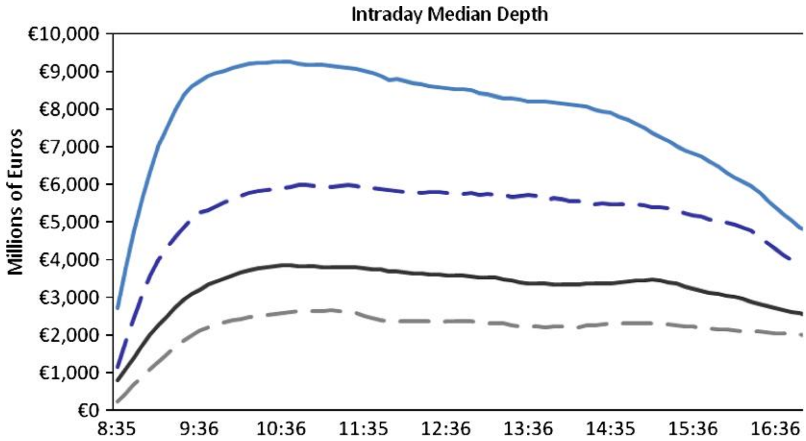

Figure 5
Intraday median depth
This figure plots the intraday median depth on e-MID. The top solid line refers to the median bid in the pre-crisis period, while the top broken line refers to the median bid in the crisis period. Similarly, the bottom broken line refers to the median ask in the crisis period, while the bottom solid line refers to the median ask in the pre-crisis period.

Notably, however, the patterns across the trading day appear to remain similar, if only slightly magnified, during the crisis. The market opens with relatively small depth, which increases dramatically over the first half hour of trading, peaks around 10:30 a.m. ECT, and declines steadily until 3:00 p.m. and sharply immediately prior to 5:00 p.m.

Figure 6 displays the number of quotes on the bid and offer side across the trading day. Quote updates on the bid side are more frequent than updates to the ask side. Interestingly, there has been only a small reduction in the number of quotes presented during the crisis, implying that the e-MID market remains an active venue for interbank liquidity, but only for diminished quantities, a clear indication of risk-averse behavior.

Figure 7 depicts the differences between the daily volume-weighted average transaction prices on e-MID and the ECB target rate. The e-MID rate is generally higher than the ECB target rate during the pre-crisis period, with periodic episodes in which e-MID rates are lower. During the crisis, however, interbank rates on e-MID are usually lower than the ECB target rate. This is the result of commercial banks' behavior during the crisis period. In fact, anecdotal evidence suggests that commercial banks decided to *front load* their reserve accounts at the beginning of the MRP, thus leading to a lower demand by the end of the MRP.[12] This behavior results in the

[12] Central banks during the crisis encouraged this behavior by providing additional funds at the beginning of the MRP (see Eisenschmidt, Hirsh, and Linzert 2009).

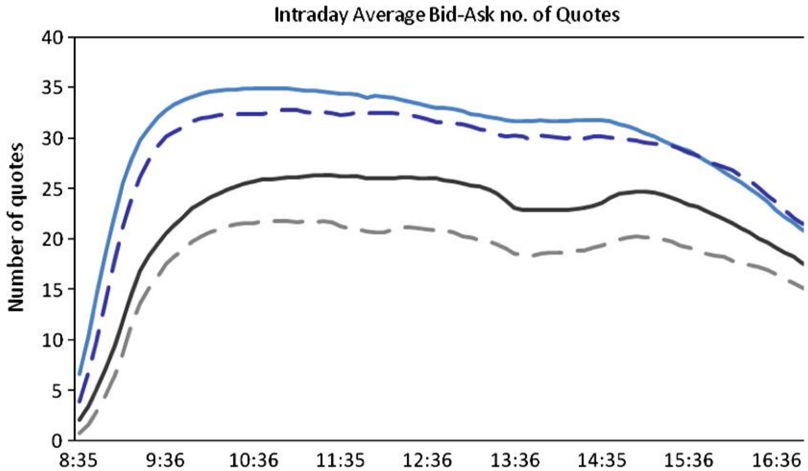

Figure 6
Intraday bid-ask number of quotes
This figure plots the intraday average number of quotes on both sides of the market posted on e-MID. The top solid line refers to the average number of bids in the pre-crisis period, while the top broken line refers to the average number of bids in the crisis period. Similarly, the bottom broken line refers to the average number of asks in the crisis period, while the bottom solid line refers to the average number of asks in the pre-crisis period.

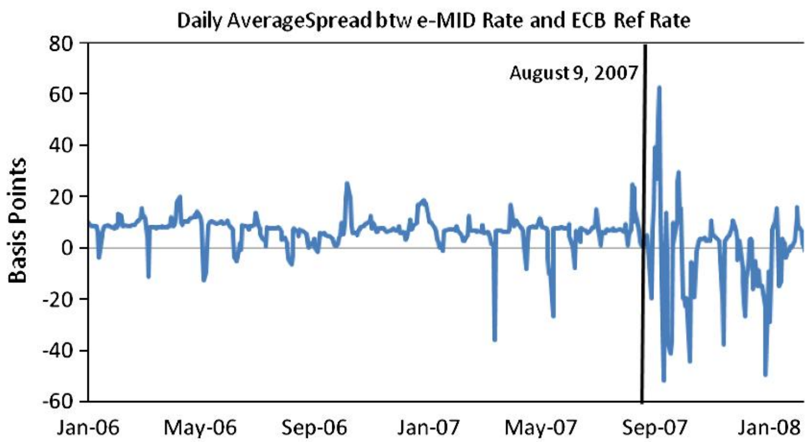

Figure 7
Daily average spread between the e-MID transaction price and ECB reference rate
This figure plots the daily average spread between e-MID transaction prices and the ECB reference rate. The vertical line indicates the beginning of the crisis period.

clear pattern of cyclicality depicted in Figure 7. Figure 7 also demonstrates that the e-MID/reference rate spread is more volatile during the crisis.

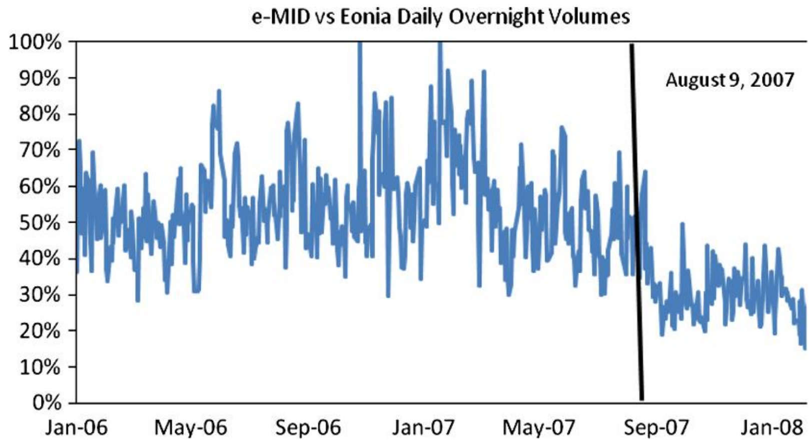

Figure 8
Daily e-MID volume as percentage of EONIA volume
This figure plots the daily overnight volume traded on e-MID as a percentage of the EONIA overnight volume. The vertical line indicates the beginning of the crisis period.

Figure 8 displays e-MID traded volume as a percentage of Euro Overnight Index Average (EONIA) volume.[13] Prior to the crisis, e-MID volume was approximately 50% of EONIA volume, dropping to 30% after August 9, 2007. In fact, during the crisis, EONIA volume increased while e-MID volume decreased, reflecting bank reluctance to reveal funding liquidity needs in the public interbank market.

Figure 9 depicts intraday average trading volume in both the pre-crisis and crisis periods. Following the market opening at 8:30 a.m., trading volume grows rapidly, peaking at 9:30 a.m. pre-crisis and somewhat later at 9:45 a.m. during the crisis. Trading activity diminishes for the next few hours, with the lowest volume periods in the early afternoon (between 1:15 and 2:15 p.m.). Volume increases thereafter, peaking again at 4:45 p.m., and falls off over the subsequent hour of trading. The reduction of trading activity during the crisis period is concentrated in the morning. During the rest of the day, there is not a significant difference between the two subperiods.

3. Econometric Approach

From the raw e-MID transaction data, we sample the overnight interest rate at 5-minute intervals. When there is no transaction in the 5-minute interval, we carry over the market price from the previous trade. Since price levels are

[13] EONIA is an effective overnight rate computed as a weighted average of all overnight unsecured lending transactions in the interbank market by a set of contributing banks. The EONIA is one of the two benchmarks for the money and capital markets in the euro zone (the other being EURIBOR). EONIA reference rates are calculated by the ECB and published by Reuters every day before 7:00 p.m. ECT.

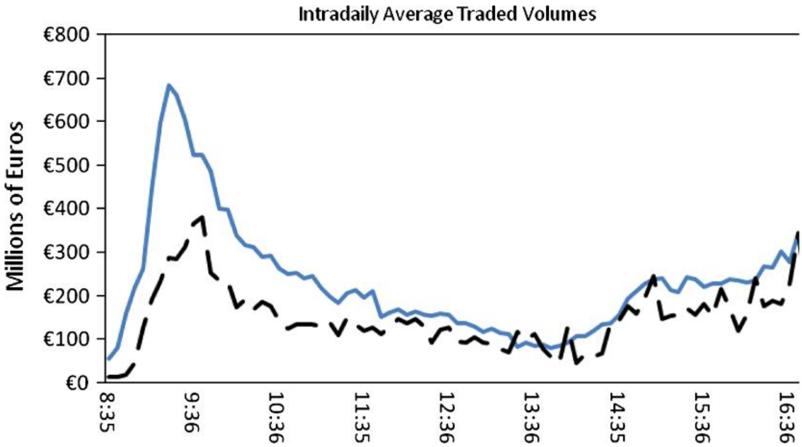

Figure 9
Intraday average e-MID traded volume
This figure plots the average intraday volume traded on e-MID. The solid line refers to the pre-crisis period, while the broken line refers to the crisis period.

non-stationary (the augmented Dickey–Fuller test fails to reject the null of a unit root), we analyze first differences in our empirical tests.

Table 1 presents summary statistics for 5-minute price changes, spreads, and trading volume. The mean price change (expressed in basis points) over the sample period is negative, reflecting a drop in the price. As indicated in Table 1, this reduction is more evident in the crisis period. This is not surprising, since reference rates worldwide (from the ECB, Fed, Bank of England, etc.) have been dropping in an attempt to provide more liquidity to markets during the crisis. Additionally, the standard deviation of the price changes is particularly high during the crisis and clearly dominates the mean during this period. This finding reflects the fact that the crisis created uncertainty about the status of the banking system—uncertainty that is reflected in higher levels of price volatility.

Bid-ask spreads also increase dramatically, reflecting higher transaction costs and lower liquidity during the crisis. The average spread during the crisis is almost four times higher than that before the crisis. The daily spread range and the volatility of the spread also increase dramatically during the crisis.

Average e-MID trading volume (in millions) decreases from £203.5 pre-crisis to £139.1 during the crisis. Increased uncertainty during the crisis appears to have increased spreads and reduced liquidity, resulting in lower trading volume. Interestingly, however, the volatility of trading volume actually fell during the crisis.

For *intervention news* events related to expected and realized macroeconomic fundamentals, we collect the exact day and time of the announcement and realization from the ECB's website. We construct the *Intervention News*

Table 1
Summary statistics: e-MID data

	?P		Spread		Volume	
	pre-crisis	Crisis	pre-crisis	crisis	pre-crisis	Crisis
Mean	−0.0197	−0.1172	2.9303	11.0400	203.50	139.09
Median	0.0000	0.0000	1.9385	6.7101	90.000	60.000
Max	70.00	70.00	77.216	131.14	6097.5	3038.6
Min	−75.000	−75.000	0.2476	0.1671	0.0000	0.0000
Std. dev.	2.6019	6.1730	4.1307	10.821	317.21	218.09
Skewness	−0.8782	−0.3594	10.108	2.6519	3.8483	3.7583
Kurtosis	221.79	26.858	137.91	12.712	29.330	25.701
ADF	−85.866	−68.407	−14.949	−11.903	−60.979	−43.980
p-value	0.0001	0.0001	0.0000	0.0000	0.0001	0.0000
ACF (1)	−0.3239	−0.3878	0.9838	0.9778	0.2370	0.2035
Q-Test (1)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ACF (10)	−0.0351	−0.0077	0.8429	0.8464	0.0327	0.0086
Q-Test (10)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ACF (50)	−0.0083	−0.0338	0.3568	0.5294	−0.0509	−0.0202
Q-Test (50)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

This table presents summary statistics of the e-MID interbank market data. ?P is the first difference in transaction prices in basis points; Spread is the 5-minute time-size weighted average spread in basis points; Volume is the 5-minute cumulative transaction volume in £ (millions); Q-Test(i) refers to the probability of the null of zero serial correlation up to lag i. Full sample: January 2, 2006–April 1, 2008; pre-crisis sample: January 2, 2006–August 8, 2007; crisis sample: August 9, 2007–April 1, 2008.

variable as the difference between these expectations and realizations, reporting each variable in Table 2.[14]

For 118 MROs, we are able to construct the following shocks, which we use as regressors. We define one supply shock as the difference between the Friday ECB announced benchmark and the quantity actually allotted. This supply shock represents the *unexpected* amount of liquidity that the ECB supplies to the system (denoted “MRO Supply Imbalance”). We further define a second shock as the difference between the amount of liquidity that the ECB supplies to the system relative to the quantity demanded by commercial banks—the MRO *cover-to-bid ratio*. This ratio captures the fraction of the quantity bid by commercial banks that is ultimately supplied and can proxy for relatively positive and negative news. When an ECB injection fully covers the quantities bid by commercial banks, the ECB conveys relatively bad news, whereas a partial filling of bids conveys relatively good news.[15]

We adopt the same approach in treating LTROs and FT operations, computing supply imbalances and cover-to-bid ratios when possible. Since we do not have LTRO announcements (benchmarks) or MRO announcements prior to November 2007, we compute no supply imbalances for these interventions.

[14] We thank ECB and Fed officials for valuable conversations for understanding and constructing these intervention variables.

[15] Drehmann, Eisenschmidt, and Nicolau (2009) use a similar approach to construct a measure of funding liquidity risk.

Table 2
ECB operations: Regressors

Name	Explanation	N pre-cr.	N crisis
Rate surprise	Reference Rate Surprise is obtained by subtracting the expected Reference rate change to the actual Reference Rate change.	20	7
MRO cover-to-bid ratio	Main Refinancing Operation (MRO) cover-to-bid ratio is the ratio between the allotted amount by the ECB (Allotment) and the quantity bid by commercial banks (Bid). It indicates the demand-supply imbalance. The duration of MROs is usually one week.	84	34
LTRO cover-to-bid ratio	Long Term Refinancing Operation (LTRO) cover-to-bid ratio is the ratio between the allotted amount by the ECB (Allotment) and the quantity bid by commercial banks (Bid). It indicates the demand-supply imbalance. The duration of LTROs is usually three months.	19	8
Special LTRO cover-to-bid ratio	Long Term Refinancing Operation (LTRO) cover-to-bid ratio is the ratio between the allotted amount by the ECB (Allotment) and the quantity bid by commercial banks (Bid) for special LTROs during the crisis. The duration of LTROs is usually three months.	0	6
FT cover-to-bid ratio	FT Bid-Allotment cover-to-bid ratio is the ratio between the allotted amount by the ECB (Allotment) and the quantity bid by commercial banks (Bid) in Fine Tuning operations. It indicates the demand-supply imbalance. The duration of FTs is usually one day.	17	7
Special FT cover-to-bid ratio	FT cover-to-bid ratio is the ratio between the allotted amount by the ECB (Allotment) and the quantity bid by commercial banks (Bid) in special Fine Tuning operations during the crisis. The duration of FTs is usually one day.	0	18
MRO supply imbalance	Main Refinancing Operation (MRO) Allotment-Benchmark is obtained subtracting the benchmark announced by the ECB (Benchmark) to the quantity actually allotted (Allotment). We have this variable only for part of the 2nd subsample (starting Nov 2007): supply side shock. The duration of MROs is usually one week.	0	22
FT supply imbalance	FT Allotment-Benchmark is obtained subtracting the benchmark announced by the ECB (Benchmark) to the quantity actually allotted (Allotment) in Fine Tuning operations: supply side shock. The duration of FTs is usually one day.	17	7
Special FT supply imbalance	FT Allotment-Benchmark Special is obtained subtracting the benchmark announced by the ECB (Benchmark) to the quantity actually allotted (Allotment) in special Fine Tuning operations during the crisis: supply side shock. The duration of FTs is usually one day.	0	18

Pre-crisis sample: January 2, 2006–August 8, 2007; crisis sample: August 9, 2007–April 1, 2008. Data are collected from the ECB official website.

Since LTROs are not usually used for daily reserve management, these shocks may be less important to the interbank market during the short intervals we study. During the crisis period, some LTRO operations appear to occur with peculiar timing, duration, and size—we denote these as *special*.

For 42 FTs, we compute supply imbalances and cover-to-bid ratios. Although FTs, by definition, are smaller in scale than MROs, these operations might be expected to have a significant impact on the interbank market because they provide funding of last resort just before the end of the MRP. We denote 18 FTs as *special* during the crisis period when they occur with peculiar timing, duration, and size.

ECB target rates are announced at the end of the monthly meeting by the ECB Monetary Policy Committee. Following Ehrmann and Fratzscher (2005a), we compile expected rates from Reuters surveys of 40 European commercial bank treasurers. From these surveys, we compute expected target rates and target rate surprises, defined as the difference between expected changes and announced changes in the target rate.

Table 3 reports summary statistics for ECB interventions. Rate surprises are negative in both periods, indicating that the average ECB rate cut exceeds market expectations. Mean (and median) rate surprises during the crisis, however, are less negative. Interestingly, the standard deviation of rate surprises does not differ between the two periods. Cover-to-bid ratios (with the exception of FTs) are generally lower during the crisis period. As expected, the volatility of the cover-to-bid ratios and supply imbalances are higher during the crisis.

Following the literature on announcements (see Andersen, Bollerslev, Diebold, and Vega 2002), we standardize our news variables. The standardized surprise associated with variable k at time t is

$$S_{k,t} = \frac{R_{k,t} - \overline{R_k}}{\sigma_k},$$

where $R_{k,t}$ is the surprise k, $\overline{R}$ is its sample average, and σ_k is the sample standard deviation. Interestingly, standardized pre-crisis rate surprises are negative but standardized crisis period surprises are positive.[16] In this regard, ECB interventions are relatively more aggressive during the crisis. Cover-to-bid ratio surprises are always positive during the crisis, indicating that banks demanded more liquidity than the ECB allotted. Conversely, pre-crisis cover-to-bid ratio surprises are negative, indicating that commercial banks demanded less liquidity than allotted. Supply imbalances are consistently negative, indicating that the central bank consistently allots less than the quantity announced on Fridays. Mean supply imbalances addressed through special operations, however, are positive.

Our econometric analysis focuses on the effect of intervention news on the e-MID interbank market. We first estimate our model for the full sample period and follow up by examining pre-crisis and crisis subperiods in order to assess the impact of the credit crunch on the interbank market under these two regimes. We estimate a model with high-frequency data that allows for the possibility of news to affect both the conditional mean and the conditional standard deviation of price changes, spreads, and trading volume. We model each variable (over 5-minute intervals) as a function of lagged values of itself and J lags of news on each of K surprises:

$$y_t = \beta_0 + \sum_{i=1}^{I} \beta_i y_{t-i} + \sum_{k=1}^{K} \sum_{j=1}^{J} \beta_{k,j} S_{k,t-j} + \varepsilon_t. \tag{1}$$

16 To conserve space, we report summary statistics for $R_{k,t}$ in Table 3 but not for $S_{k,t}$.

Table 3
Summary statistics: ECB interventions, $R_{k,t}$

	mean		*median*		*max*		*min*		*std. dev.*		*skewness*		*kurtosis*	
	pre-crisis	*crisis*	*pre-crisis*	*crisis*	*pre-crisis*	*crisis*	*pre-crisis*	*crisis*	*pre-crisis*	*crisis*	*pre-crisis*	*crisis*	*pre-crisis*	*crisis*
Rate surprise	−1.7631	−1.2449	−2.5240	−0.5986	4.9548	0.9965	−5.7958	−8.2195	3.0007	3.1888	0.7805	−2.2613	−0.4020	5.5216
MRO cover-to-bid ratio	0.8044	0.6547	0.7952	0.6551	0.9982	0.9243	0.6965	0.4197	0.0619	0.1170	0.6604	0.0112	0.3170	−0.2030
LTRO cover-to-bid ratio	0.6649	0.5372	0.6536	0.4827	0.8032	1.0000	0.5394	0.3777	0.0724	0.2033	0.1571	2.0196	−0.3370	4.6076
Special LTRO cover-to-bid ratio		0.4715		0.4960		0.5707		0.3180		0.0978		−0.7237		−0.7573
FT cover-to-bid ratio	0.5926	0.6558	0.5469	0.6089	1.0000	1.0000	0.0761	0.1996	0.3555	0.2901	−0.1410	−0.3578	−1.7094	−1.0116
Special FT cover-to-bid ratio		0.7338		0.9197		1.0000		0.1675		0.3116		−0.7519		−1.0121
MRO supply imbalance		35.505		20.000		217.00		3.500		52.817		2.8817		7.9264
FT supply imbalance	−0.7782	−0.536	0.0000	0.0000	2.5000	5.5000	−8.2000	−9.2500	2.4285	4.3551	−2.2219	−1.2460	5.4832	3.8025
Special FT supply imbalance		11.175		0.0000		94.841		−48.420		33.988		0.7484		1.0858

All variable definitions are identical to Table 2; rate surprise in basis points; cover-to-bid ratios are bounded between zero (the ECB allots no liquidity to banks) and one (full allotment); supply imbalances are in £ (billions); pre-crisis sample: January 2, 2006–August 8, 2007; crisis sample: August 9, 2007–April 1, 2008.

We allow the disturbance term in the conditional mean equation to be heteroscedastic. Following Andersen and Bollerslev (1998) and Andersen, Bollerslev, Diebold, and Vega (2002, 2007), we estimate the model using a two-step weighted least squares procedure. We first estimate the conditional mean model (1) by ordinary least squares (OLS) regression, and then we estimate the time-varying volatility of ε_t from the regression residuals, which we use to perform a weighted least squares estimation of Equation (1). We approximate the disturbance volatility using the following specification:

$$\left|\hat{\varepsilon}_t\right| = c + \psi \frac{\hat{r}_{d(t)}}{\sqrt{n}} + \sum_{k=1}^{K} \sum_{j'=1}^{J'} \beta_{kj'} \left|S_{k,t-j'}\right| + \left\{ \sum_{q=1}^{Q} \left[\delta_q \cos\left(\frac{2q\pi t}{n}\right) + \varphi_q \sin\left(\frac{2q\pi t}{n}\right) \right] \right\} + u_t. \tag{2}$$

The left-hand-side variable $\left|\hat{\varepsilon}_t\right|$ is the absolute value of the residual of Equation (1), which proxies for the volatility in the 5-minute interval t. We model the 5-minute volatility as driven (i) partly by the volatility over the day containing the 5-minute interval in question, which is represented by $r_{d(t)}$; (ii) partly by the shock $S_{k,t}$; and (iii) partly by a calendar component consisting largely of intraday effects (seasonality) that capture the high-frequency deviations of intraday volatility from the daily average.[17] We denote $r_{d(t)}$ as the one-day-ahead volatility forecast for day $d(t)$ (the day that contains interval t) and compute the volatility forecast as follows. First, we compute an efficient measure of daily volatility using the range (see Brandt and Diebold 2006) that is equal to

$$r_{d(t)} = \sqrt{\frac{\pi}{8}} * \ln\left[\frac{\max\left(y_{d(t)}\right)}{\min\left(y_{d(t)}\right)}\right],$$

where y is the set of observations belonging to $d(t)$. We then estimate the OLS model ($r_{d(t)} = c + \beta r_{d(t-1)} + \varepsilon_t$) and compute the one-day-ahead volatility forecast.

4. Results

Models (1) and (2) provide an accurate approximation of the dynamics of both the conditional mean and the conditional standard deviation of the first difference in transaction prices.[18] For the sake of brevity, we do not present the

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17 We represent the calendar component by a Fourier flexible form with trigonometric terms that have a periodicity of one day and where n is equal to the number of 5-minute intervals.

18 Using Akaike and Schwartz information criteria, we select an optimal lag length of 10 for the autoregressive component in Equation (1) for the pre-crisis period and 7 during the crisis. The optimal lag length for $S_{k,t}$ in the conditional mean equation is 6 for the pre-crisis period and 2 for the crisis period, implying that interventions are absorbed into interbank prices very quickly. Complete results are available from the authors upon request.

numerous coefficients in our model, but interbank prices typically react significantly to liquidity shocks. Most pre-crisis coefficients are only marginally statistically significant, in contrast to the crisis period, in which central bank interventions are statistically more important.

We also obtain the specification of the variance equation by minimizing Akaike and Schwartz information criteria.[19] In line with the findings for the conditional mean equation, price volatility also adjusts quickly to ECB interventions. All shocks, $S_{k,t}$, are statistically significant in the standard-deviation equation in every sample considered. Interestingly, most coefficients are positive, indicating that volatility increases with central bank intervention, particularly during the crisis period. To illustrate, Figure 10 shows the estimated parameters for the pre-crisis and the crisis periods for MRO cover-to-bid ratios. In the pre-crisis period, MRO cover-to-bid ratios moderately increase price volatility, but the effect dies out quickly. During the crisis, however, MRO cover-to-bid ratios have a much stronger and more persistent impact on volatility. Higher volatility levels due to central bank intervention indicate that these interventions increase the overall uncertainty, but also convey information to the e-MID market.

We also apply models (1) and (2) to the dynamics of ECB interventions on e-MID interbank spreads and spread volatility.[20] Interestingly, we find that

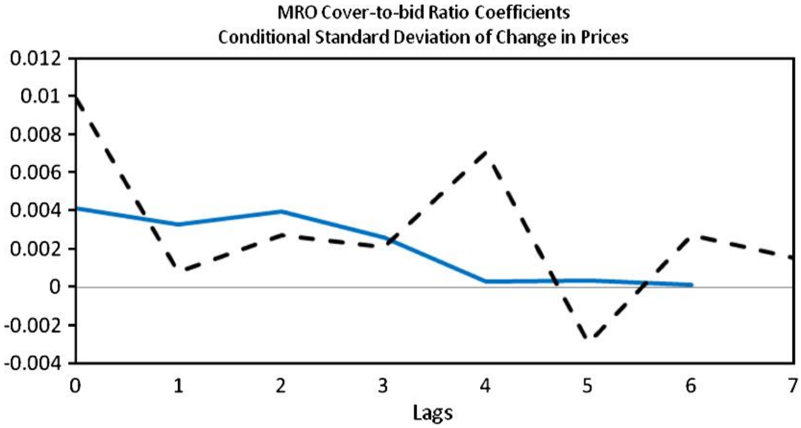

Figure 10
Parameter estimates of the MRO cover-to-bid ratio in Equation (2) for price changes
This figure plots estimates of the interbank market reaction to MROs, from Equation (2), for price changes with the MRO cover-to-bid ratio as regressor. The solid line refers to the pre-crisis period, while the broken line refers to the crisis period.

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19 The optimal specification for the Fourier polynomial is $Q = 5$ for the pre-crisis period and $Q = 7$ for the crisis period. The optimal lag length for the central bank interventions is 6 for the pre-crisis period and 7 for the crisis period.

20 For these estimations, the optimal specification for the autoregressive components is 12 for the pre-crisis period and 3 for the crisis period. The optimal lag length for $S_{k,t}$ in these equations is 7 for the pre-crisis period and 2 for the crisis period.

general spread reductions found during the pre-crisis periods do not extend to the crisis period. In fact, during the crisis period, interventions have little effect or appear to increase conditional interbank market spreads. Given that one motivation for ECB intervention is to improve liquidity in the interbank market, this result stands out as notable. We interpret this evidence as suggesting that ECB interventions during the crisis fail to mitigate asymmetric information in the interbank lending market.

It is perhaps unsurprising, given the breadth and the amplitude of the crisis, that the interbank market did not react favorably to monetary policy. Neither the uncertainty about counterparty risk nor idiosyncratic risk is reduced by central bank intervention, and in fact, financial institutions may be predisposed to hoard capital during the crisis (Heider, Hoerova, and Holthausen 2009). Nevertheless, interventions are quickly embedded in the volatility of the spread.[21] Moreover, central bank operations consistently increase the volatility of interbank spreads, particularly during short-term interventions. Figure 11 shows coefficients for FT cover-to-bid ratios before and during the crisis. Pre-crisis FT cover-to-bid ratios moderately increase volatility at first, but volatility is mean reverting at lags three and four. This is not the case during the crisis period, when FT cover-to-bid ratios always increase spread volatility. These results

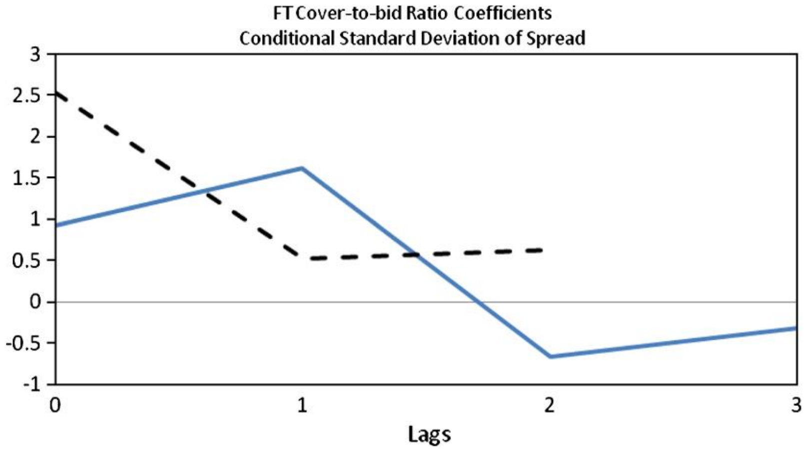

Figure 11
Parameter estimates of the FT cover-to-bid ratio in Equation (2) for spreads
This figure plots estimates of the interbank market reaction to FT operations, from Equation (2) for spreads with the FT cover-to-bid ratio as regressor. The solid line refers to the pre-crisis period, while the broken line refers to the crisis period.

[21] In Equation (2), $Q = 24$, 15 and $j = 3$, 2 in the pre-crisis and crisis periods, respectively.

indicate that one main objective of central bank intervention—to reduce liquidity risk—is not attained during the crisis.[22]

Similar to spreads and price changes, policy shocks are quickly incorporated into volume.[23] Pre-crisis interventions generally enhance interbank trading volume. During the crisis, however, trading volume is largely unresponsive to or declining with central bank interventions. This result suggests that the increased spreads and volatility may stem from ECB participation crowding out trades that may have otherwise occurred in the interbank market. This effect is particularly magnified for short-term interventions. These results are consistent with the possibility that commercial banks substitute liquidity from the central bank during the intervention and therefore do not need to trade on e-MID. Figure 12 shows that unexpected changes in rates actually reduce the volatility of volume pre-crisis, but increase the volatility of volume during the crisis.

Our analysis is carried out using high-frequency data, and we obtain estimates from observations at 5-minute intervals. Although we measure short-term market reactions (within minutes) to policy shocks set a day (or days) prior to executed interventions, the central bank takes into account market conditions when intervening, creating potential endogeneity issues. We explore potential endogeneity biases formally following Forbes and Rigobon (2002), who demonstrate that the bias in a linear regression due to differing

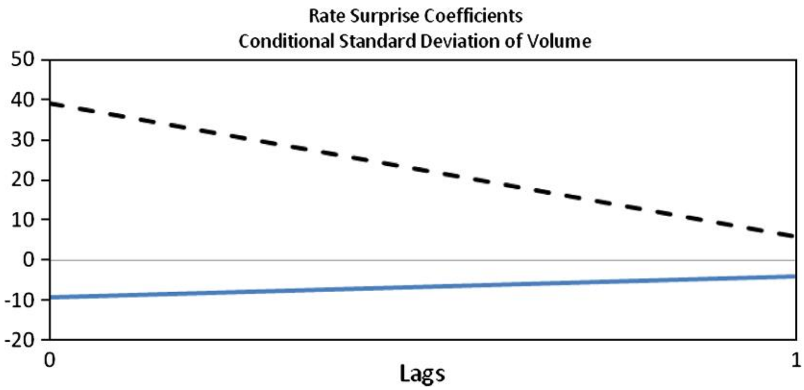

Figure 12
Parameter estimates of the rate surprise in Equation (2) for volume
This figure plots estimates of the interbank market reaction to rate surprises, from Equation (2) for volume with the rate surprise as regressor. The solid line refers to the pre-crisis period, while the broken line refers to the crisis period.

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22 "The TAF (Term Auction Facility) is a tool to reduce liquidity risk in the market by improving the allocation of funds to depositary institutions" (McAndrews, Sarkar, and Wang, 2008 p.5)

23 The optimal lag length in the conditional mean and standard deviation of volume is one. We include seven autoregressive lags in the conditional mean for the full sample and pre-crisis and six lags during the crisis.

volatilities across periods accounts for both omitted variable and endogeneity biases. More formally in our setting, the bias is given by

$$b = \frac{\sigma_{xx}^{c}}{\sigma_{xx}^{p}} - 1,$$

where σ_{xx}^{c} is the variance of x in the crisis period and σ_{xx}^{p} is the variance of x in the pre-crisis period. For all nine ECB intervention types described in Table 2, this bias is always nearly identically zero, suggesting that our estimates are free from endogeneity and omitted variable biases.

5. Impact Responses

For a richer description of the effects of central bank policies before and during the crisis, we estimate impact-response functions. To focus on the impact of central bank shocks, we estimate the model

$$y_t = \beta_{1,k} S_{k,t} + \beta_{2,k} D_{k,t} S_{k,t} + \varepsilon_t \tag{3}$$

(see Andersen, Bollerslev, Diebold, and Vega 2002), where the yt variables represent the change in price, spread, and trading volume at 5-minute intervals, Sk, t is the standardized central bank policy ($k = 9$ different intervention types described in Table 2), and Dk, t is a dummy variable that takes the value one in the crisis period. We estimate Equation (3) over the full sample period, utilizing the dummy variable for identifying incremental effects during the crisis. However, since four of the nine intervention types occur only during the crisis, we estimate the model separately for these cases. To account for effects on the second moment, we also estimate Equation (3) for the absolute value of yt.

The parameters $\beta 1$ and $\beta 2$ can be interpreted as an impact response of yt to central bank policies. Results for price changes are reported in Table 4. We find that at these short horizons, pre-crisis interventions generally do not affect prices, while crisis interventions have many differing effects.

More consistently, ECB interventions persistently increase price volatility in both pre-crisis and crisis periods, with a much stronger marginal increase during the crisis period. Five of our eight interventions (including all MROs) appear to create significantly greater volatility during the crisis. These volatility changes also may suggest that interventions are important information events.

Table 5 reports estimation results of Equation (3) for quoted spreads. ECB interventions largely reduce spreads in the pre-crisis period, enhancing interbank market liquidity. However, interventions during the crisis generally increase spreads. This inconsistency suggests that the underlying conditions during the crisis period differ from pre-crisis conditions. ECB interventions also increase the volatility of spreads (and more significantly during the crisis),

Table 4
Impact responses: Change in prices

	Conditional Mean			Conditional St. Dev.		
$S_{k,t}$	Pre-crisis (β_1)	Crisis ($\beta_1+\beta_2$)	R^2	Pre-crisis (β_1)	Crisis ($\beta_1+\beta_2$)	R^2
Rate surprise	0.1015	−0.0531	0.0014	0.2344	1.8824*	0.0688
	(0.0776)	(0.3101)		(0.3630)	(0.9152)	
MRO cover-to-bid ratio	−0.0115	1.1443	0.0094	0.1324	2.0874*	0.0374
	(0.4982)	(1.5799)		(0.6963)	(1.2306)	
LTRO cover-to-bid ratio	1.3895	−1.2815	0.1114	0.2770	2.6210	0.2038
	(1.8962)	(3.5537)		(1.3723)	(2.6321)	
FT supply imbalance	76.977*	65.821*	0.6159	81.069*	88.150*	0.5965
	(34.479)	(35.6713)		(39.460)	(44.0639)	
FT cover-to-bid ratio	5.7263	−6.6007*	0.2181	7.6378	15.644	0.1190
	(4.8717)	(3.9024)		(8.3773)	(17.055)	
MRO supply imbalance		0.7410*	0.0163		1.4780*	0.0296
		(0.3649)			(0.6457)	
Special LTRO cover-to-bid ratio		−0.9210*	0.0747		1.0229	0.0139
		(0.0623)			(0.6372)	
Special FT supply imbalance		4.1158	0.1230		6.0630	0.0209
		(4.4609)			(4.5358)	
Special FT cover-to-bid ratio		3.3830*	0.0632		6.0310*	0.0035
		(0.3320)			(3.1031)	

This table presents impulse responses for the change in price at five-minute intervals. $S_{k,t}$ is the standardized central bank policy. All variable definitions are identical to Table 2. Newey–West standard errors in parentheses; β_x refers to the pre-crisis period: January 2, 2006–August 8, 2007; $\beta_x+\beta_y$ refers to the crisis period: August 9, 2007–April 1, 2008. * indicates significance at 10% level or higher.

Table 5
Impact responses: Spreads

	Conditional Mean			Conditional Variance		
$S_{k,t}$	Pre-crisis (β_1)	Crisis ($\beta_1+\beta_2$)	R^2	Pre-crisis (β_1)	Crisis ($\beta_1+\beta_2$)	R^2
Rate surprise	−0.8996*	2.8885*	0.3117	1.0525*	4.7652*	0.2701
	(0.3446)	(1.1765)		(0.6331)	(1.1543)	
MRO cover-to-bid ratio	1.0176	7.6529*	0.1675	0.9726	7.8392*	0.1595
	(0.7746)	(3.0603)		(1.4007)	(3.8585)	
LTRO cover-to-bid ratio	−3.0614	7.1486*	0.6311	2.6684	8.2689*	0.5191
	(1.8977)	(4.0834)		(1.8636)	(3.6442)	
FT supply imbalance	−137.71*	131.73	0.8328	152.08*	156.08*	0.9041
	(22.074)	(26.937)		(12.753)	(16.352)	
FT cover-to-bid ratio	−13.770*	−6.0306	0.2505	23.303*	27.361	0.1688
	(5.7944)	(13.0442)		(12.601)	(22.378)	
MRO supply imbalance		5.0270*	0.1324		9.5478*	0.0625
		(2.2944)			(1.4103)	
Special LTRO cover-to-bid ratio		3.3115*	0.3363		3.3115*	0.3363
		(0.9639)			(0.9639)	
Special FT supply imbalance		0.5421	0.0851		12.188*	0.2154
		(5.1152)			(2.7073)	
Special FT cover-to-bid ratio		−1.0171	0.1899		11.912*	0.0501
		(3.8685)			(2.8900)	

This table presents impulse responses for spreads at five-minute intervals. $S_{k,t}$ is the standardized central bank policy. All variable definitions are identical to Table 2. Newey–West standard errors in parentheses; β_x refers to the pre-crisis period: January 2, 2006–August 8, 2007; $\beta_x+\beta_y$ refers to the crisis period: August 9, 2007–April 1, 2008. * indicates significance at 10% level or higher.

Table 6
Impact responses: Volume

	Conditional Mean			Conditional St. Dev.		
$S_{k,t}$	Pre-crisis (β_1)	Crisis ($\beta_1+\beta_2$)	R^2	Pre-crisis (β_1)	Crisis ($\beta_1+\beta_2$)	R^2
Rate surprise	−0.9636	7.0792	0.0003	10.453	19.126	0.0009
	(10.784)	(33.476)		(21.216)	(33.846)	
MRO cover-to-bid ratio	30.785*	−123.97*	0.0388	−7.3214	−134.74	0.0316
	(15.680)	(66.473)		(43.515)	(94.922)	
LTRO cover-to-bid ratio	38.891	−132.41*	0.0538	−22.761	−139.52	0.0470
	(50.739)	(78.929)		(79.816)	(116.25)	
FT supply imbalance	716.90	662.47	0.0670	−848.38	−892.35	0.0828
	(618.69)	(669.14)		(670.98)	(853.68)	
FT cover-to-bid ratio	90.327	22.337	0.0328	−359.08	−398.93	0.1106
	(84.136)	(219.57)		(333.53)	(404.97)	
MRO supply imbalance		6.9191	0.0025		71.173*	0.0005
		(28.593)			(19.603)	
Special LTRO cover-to-bid ratio		33.891*	0.4520		33.891*	0.4520
		(16.912)			(16.912)	
Special FT supply imbalance		−82.979*	0.2645		117.05*	0.1103
		(39.588)			(28.050)	
Special FT cover-to-bid ratio		29.579	0.0005		125.02*	0.0015
		(38.922)			(35.005)	

This table presents impulse responses for volume at five-minute intervals. $S_{k,t}$ is the standardized central bank policy. All variable definitions are identical to Table 2. Newey–West standard errors in parentheses; β_x refers to the pre-crisis period: January 2, 2006–August 8, 2007; $\beta_x+\beta_y$ refers to the crisis period: August 9, 2007–April 1, 2008. * indicates significance at 10% level or higher.

highlighting that ECB liquidity operations exacerbate uncertainty and add to liquidity risk in the interbank market.

Table 6 reports estimation results of Equation (3) for trading volume in the interbank market. In the pre-crisis period, only the MRO cover-to-bid ratio significantly increases trading volume. Volume significantly falls for many interventions during the crisis period, however, providing further evidence that liquidity suffers during these interventions. While the volatility of pre-crisis volume does not significantly change, the volatility of volume during the crisis increases for all special interventions (as well as for MRO supply imbalances).

6. Do Interventions Simply Convey Bad News During the Crisis?

Given the evidence that economic conditions are unique during the crisis and our evidence that interventions convey information, we attempt to disentangle the information effects from liquidity effects. We first explore interbank market reactions to other news events to gauge how general news affects the market in the pre-crisis and crisis periods. We then use cover-to-bid ratios to explore the interbank reaction to relatively good and bad news revealed by interventions.

We first examine news events unrelated to interventions but having similar effects in the interbank market. To identify these, we choose 236 five-minute intervals in which price changes, spreads, and volume exceed the averages found during interventions but do not occur on intervention days. We confirm that the 179 pre-crisis events and 57 crisis events indeed correspond with important economic and political news in Factiva, Lexis, and Westlaw.[24]

Similar to ECB interventions, and perhaps not surprisingly, other news events affect the interbank market during both pre-crisis and crisis periods. Comparisons between interventions and other news events during the crisis suggest that interventions have somewhat differing effects in the interbank market. Other news events, on average, do not significantly increase spreads or reduce volume. Notably, however, other news events reduce the volatility of interbank spreads and volume, while interventions more consistently increase volatility. Although not conclusive, this evidence suggests that interventions do not simply convey information to the interbank market—other effects likely intervene.

If the ECB has important economic information that individual banks do not have, then the ECB can use interventions to reveal this information. In fact, each intervention is known well in advance, with banks bidding for funds in each operation. The ECB then chooses the level of intervention necessary after bids are received. We therefore use the cover-to-bid ratio as a proxy for good news and bad news for each intervention. A cover-to-bid ratio of 1.0 indicates that the ECB fills all bids, and suggests that the central bank agrees that each individual bank requires full liquidity injections. The ECB conveys relatively more optimistic news when choosing a lower cover-to-bid ratio, since the central bank chooses to cover fewer expressed liquidity needs in the market.

Figure 13 shows that the cover-to-bid ratio varies substantially for all interventions across our full sample period, suggesting that the ECB chooses to fill a wide range of bids, depending on economic needs and circumstances. Prior

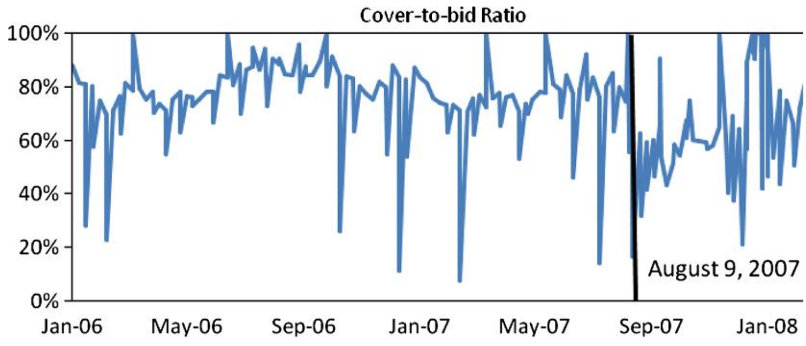

Figure 13
Cover-to-bid ratio of ECB interventions
This figure plots the ratio between amount of liquidity bid by commercial banks and amount of liquidity actually allotted by the ECB. The vertical line indicates the beginning of the crisis period.

24 Our procedure identifies the Dow Jones Index at 11,000 on January 11, 2006; a number of production announcements from the Organization of Petroleum Exporting Countries; U.S. nonfarm payroll and producer price index announcements; oil prices at $80, $90, and $100/barrel; large changes in eurozone stock markets; large shifts in £/$ exchange rates; and European firm-specific news like Deutsche Bank buying MortgageIT Holdings.

to the crisis, the ratio varies between 0.08 and 1.00. Similarly, during the crisis period, the cover-to-bid ratio varies from 0.17 to 1.00, suggesting that the central bank continues to attempt to convey relatively good and bad news during the crisis. We classify interventions as relatively good news when the cover-to-bid ratio is below the sample mean and as bad news otherwise, calculating means for the pre-crisis and crisis periods separately.[25]

Table 7 provides evidence that the cover-to-bid ratio proxies for relatively good and bad news during the pre-crisis period. Good news (relatively low cover-to-bid ratio), for instance, reduces spreads and increases volume, on average. Bad news also enhances trading volume but does not significantly affect spreads. Interventions during the pre-crisis period largely increase the volatility of each variable (except good news events, which reduce the volatility of trading volume).

During the crisis, however, the interbank market reaction to interventions is quite different, with both good and bad news interventions increasing prices and the volatilities of prices and spreads. The link between cover-to-bid ratios and good and bad news appears to be broken during the crisis. This evidence confirms that interventions generally fail to reduce prices but consistently increase volatility after controlling for relatively good and bad news during the crisis.

Of course the link may be broken because there is simply no good news to reveal during the crisis. If this were the case, however, it might be reasonable to expect that cover-to-bid ratios during the crisis might exceed pre-crisis ratios as the central bank floods the market with funds. However, crisis period cover-to-bid ratios are actually marginally significantly *lower* (at the 10% level) than pre-crisis ratios, a result that argues that either the ECB had no information advantage or some other effect intervenes. Whether this effect includes liquidity hoarding, crowding out, or another market response to interventions, our results speak to the ineffectiveness of standard intervention mechanisms to improve market liquidity during the crisis.

Indeed, the existence of special interventions suggests that ECB officials recognize that it was not “business as usual” during the crisis. The optimal experimental design would examine what would have happened if these special interventions were not implemented. In the absence of this perfect design, our results speak to this question in other ways. When we examine shocks related to typical (not special) operations, we can compare pre-crisis and crisis results on a *ceteris paribus* basis, since these operations would have occurred under normal circumstances. From this comparison, we see that liquidity is largely enhanced with pre-crisis operations but more commonly diminished during the crisis.

[25] For robustness, we also classify good news and bad news as the bottom and top quartiles of interventions and find similar results.

Table 7
Estimation results: Good and bad news

	Conditional mean – Equation (1)					
	ΔPrice		Spread		Volume	
$S_{k,t}$	Pre-crisis	Crisis	Pre-crisis	Crisis	Pre-crisis	Crisis
Bad news	−1.2785	0.1682	0.6855	2.2215	212.95∗	50.515
	(0.9685)	(0.7640)	(3.6795)	(2.0614)	(93.759)	(45.033)
Bad news (−1)	−0.3522	1.4573∗	1.2451	0.6284	−34.169	−8.4780
	(0.5616)	(0.7141)	(4.3848)	(0.6867)	(30.935)	(26.094)
Bad news (−2)	−0.0968	−0.3639				
	(0.4371)	(0.8441)				
Good news	−1.0727∗	0.0941	−0.0946∗	−0.0563	−49.532∗	10.500
	(0.5960)	(1.1827)	(0.0306)	(0.1971)	(28.480)	(35.674)
Good news (−1)	−0.7221	0.5592	0.4249	−0.3403	92.015∗	−3.5695
	(0.7462)	(1.2602)	(2.7250)	(0.4342)	(7.6891)	(20.865)
Good news (−2)	1.5296∗	4.1218∗				
	(0.8475)	(2.3314)				
	Conditional standard deviation – Equation (2)					
	ΔPrice		Spread		Volume	
Bad news	1.7153∗	0.2061	1.7653∗	4.4087∗	248.93∗	44.725
	(0.2811)	(0.8170)	(0.0888)	(0.3803)	(29.447)	(29.653)
Bad news (−1)	0.5754∗	5.1563∗	2.0501∗	1.0142∗	−4.2051	−25.400
	(0.2816)	(0.8192)	(0.0888)	(0.3808)	(29.439)	(29.672)
Bad news (−2)	0.3220	0.1200	0.1286	1.1101∗		
	(0.2816)	(0.8196)	(0.0888)	(0.3812)		
Bad news (−3)	0.1469	1.3053	0.2666∗	0.3093		
	(0.2810)	(0.8193)	(0.0888)	(0.3812)		
Bad news (−4)		0.2301	0.1518∗	0.3189		
		(0.8175)	(0.0888)	(0.3808)		
Bad news (−5)			0.1817∗	0.3435		
			(0.0888)	(0.3804)		
Good news	0.7105∗	0.9966	−0.0169	0.0930	−21.041	14.810
	(0.3005)	(0.7577)	(0.0958)	(0.3512)	(31.753)	(27.521)
Good news (−1)	1.0176∗	1.2066	1.3565∗	0.4363	−98.823∗	−21.782
	(0.3035)	(0.7576)	(0.0965)	(0.3513)	(31.760)	(27.503)
Good news (−2)	1.3442∗	−0.1322	0.8603∗	0.1264		
	(0.3035)	(0.7577)	(0.0966)	(0.3515)		
Good news (−3)	0.7740∗	−0.2179	0.0500	0.4643		
	(0.3006)	(0.7575)	(0.0966)	(0.3515)		
Good news (−4)		1.3404∗	0.3041∗	0.2868		
		(0.7572)	(0.0965)	(0.3512)		
Good news (−5)			0.1886∗	0.7274∗		
			(0.0958)	(0.3511)		

This table presents regression coefficients for the interbank market reaction to good and bad news, defined as the standardized cover-to-bid ratio of pooled intervention types. Good (bad) news is defined as liquidity injections that fall below (above) the subperiod sample mean, suggesting that the ECB injects relatively less (more) liquidity than aggregate bank bids. * indicates significance at 10% level or higher; standard errors in parentheses; pre-crisis sample: January 2, 2006–August 8, 2007; crisis sample: August 9, 2007–April 1, 2008.

7. Conclusion

Beginning in August 2007, the subprime crisis created serious liquidity issues for the interbank market that prompted multiple and varied responses from central banks attempting to improve market conditions. In this article, we employ a unique dataset of trades and quotes from an electronic interbank market in

Europe (e-MID) to gauge the effects of ECB interventions on interbank market quality from January 2, 2006, until April 1, 2008. We examine both ordinary and special interventions and find that ECB interventions quickly affect both the conditional mean and conditional standard deviation of interbank prices, spreads, and volume. Although short-term central bank policies reliably improve liquidity in the interbank market prior to the crisis, this link is broken during the crisis period. In fact, although central bank interventions are sometimes informative in the sense that they affect market prices, interventions more consistently increase spreads and other measures of market volatility during the crisis.

Central bank interventions commonly increase volatility in the market characteristics that we study, more often during the crisis. We document a breakdown between the interbank market and the cover-to-bid ratio during the crisis as well. This suggests that central bank policies may not root out interbank asymmetric information, but rather crowd out the private supply of liquidity. Interventions during the crisis commonly increase spreads and reduce trading volume, suggesting that ECB operations impede the important information aggregation, price discovery, and peer-monitoring role that interbank markets provide.

Our results suggest that ECB interventions may convey news to the interbank market, but the news is not simply bad news during the crisis. Indeed, the ECB commonly covers only a fraction of bank bids, with an even smaller average cover-to-bid ratio during the crisis. Although the pre-crisis variation in the cover-to-bid ratio distinguishes relatively good news from bad news, the interbank market does not reflect the same distinction during the crisis. Moreover, we find that ECB interventions fail to consistently mitigate asymmetric information in the interbank market.

Indeed, other anecdotal evidence also suggests that interventions during the crisis did not enhance liquidity as intended. First and foremost, the crisis appears to be persistent. In fact, since October 2008 the ECB has made at least six additional adjustments to interventions to enhance liquidity outcomes, has added special interventions, and now has begun to identify stress test results for individual banks (which more directly addresses information asymmetries between banks).

Our findings have important policy implications about how central banks address liquidity crises. The limited ability of direct liquidity operations to improve market liquidity suggests that central banks should consider additional tools to mitigate asymmetric information during periods of crisis. Releasing individual bank results from stress tests is a step in the right direction. The clear implication from our analysis is that central bank actions that target the root cause of the crisis are preferred to more routine, or even more frequent, liquidity injections. While a complete analysis of these alternatives is beyond the scope of the current article, we see these as fruitful research topics in the future.

Appendix: Computation of the Spread

We define the spread d_t at time t as the difference between the ask and bid quotes, $d_t = a_t - b_t$. If more than one quote is active at the same time, the average spread weighted by the size of each quote i is

$$d_t = \frac{\sum_{i=1}^{N} Q^i_{a,t} a^i_t}{\sum_{i=1}^{N} Q^i_{a,t}} - \frac{\sum_{i=1}^{M} Q^i_{b,t} b^i_t}{\sum_{i=1}^{M} Q^i_{a,t}},$$

where a^i_t is the ask-quote i active at time t, $Q^i_{a,t}$ is the size of ask-quote i active at time t, b^i_t is the bid-quote i active at time t, and $Q^i_{b,t}$ is the size of bid-quote i active at time t. By definition, d_t is active between time t and $t + dt$; therefore, a time-weighted measure of the already weighted-by-size spread is

$$d\,(t_0, t_0 + \Delta t) = \frac{1}{\Delta t} \int_{t_0}^{t_0+\Delta t} d_t * dt.$$

In discrete time, each 5-minute interval has a number K of finite subintervals in which we compute the average spread weighted by the size of each quote. The interval k is identified as the *minimum* time interval during which at least one quote is active. Each k subinterval contains i ask-quotes and j bid-quotes. The spread for the subinterval k is

$$d_k = \frac{\sum_{i=1}^{N} Q^i_{a,k} a^i_k}{\sum_{i=1}^{N} Q^i_{a,k}} - \frac{\sum_{i=1}^{M} Q^i_{b,k} b^i_k}{\sum_{i=1}^{M} Q^i_{a,k}}.$$

We then compute the spread d_h used as a reference for interval h as the time-weighted average of the subinterval spreads $d_k$belonging to interval h. If Δk is the duration of each interval k, then a time-weighted measure of the already weighted-by-size spread for the interval h is

$$d_h = \frac{\sum_{k=1}^{K} d_k \, \Delta k}{\sum_{k=1}^{K} \Delta k} = \frac{\sum_{k=1}^{K} d_k \, \Delta k}{300}.$$

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